

Risk Stratification Scores in Anaesthesia & Perioperative Medicine

A system-wise and age-group-wise reference, with notes on validation in the Indian population

Compiled July 2026. Scores are listed once, in the category where they are most commonly used. Where a score spans categories (e.g. ASA), it is placed under its primary home and cross-referenced rather than repeated.

This document lists only genuine **risk-stratification scores** — tools that take patient/operative variables and output a predicted probability or risk class for a surgical/anaesthetic population. Pure functional/symptom classifications (e.g. NYHA, CCS, METs, ECOG) and ICU/sepsis/deterioration tools (e.g. APACHE, SOFA, qSOFA, NEWS) are excluded by design.

PART A — CLASSIFIED BY ORGAN SYSTEM

1. General / Global Perioperative Risk (whole-patient, mortality & morbidity)

Score	Predicts	Core inputs
ASA Physical Status Classification (ASA-PS I-VI, +E)	Overall perioperative risk / baseline health	Systemic disease burden, functional limitation
Surgical Apgar Score (SAS)	30-day major morbidity/ mortality	Intra-op estimated blood loss, lowest MAP, lowest HR
POSSUM (Physiological & Operative Severity Score for enUmeration of Mortality & morbidity)	Morbidity & mortality	12 physiological + 6 operative variables
P-POSSUM (Portsmouth modification)	Mortality (better calibrated than POSSUM)	Same variables, revised equation
Variant POSSUMs — Cr-POSSUM (colorectal), V-POSSUM/V-P-POSSUM (vascular), O-POSSUM (oesophagogastric), J-POSSUM (Jabalpur, India)	Specialty-specific mortality	Adapted POSSUM equations
SORT (Surgical Outcome Risk Tool) & SORT v2	30-day mortality	ASA, urgency, severity, specialty, cancer, age
ACS-NSQIP Universal Surgical Risk Calculator	30-day complications, mortality, LOS	21 patient + procedure (CPT) variables
Charlson Comorbidity Index (CCI)	1- and 10-year comorbidity-adjusted mortality	Weighted comorbidities + age
POSPOM (Preoperative Score to Predict Postoperative Mortality)	In-hospital mortality	Age, comorbidities, procedure type
E-PASS (Estimation of Physiologic Ability & Surgical Stress)	Postoperative morbidity/ mortality	Preoperative risk score + surgical stress score
Surgical Risk Scale (SRS, Sutton)	Mortality across risk spectrum	CEPOD urgency + ASA + BUPA operative severity (score 3–14)
S-MPM (Surgical Mortality Probability Model)	30-day mortality (non-cardiac)	3 factors: ASA, emergency status, surgery risk class
SURPAS (Surgical Risk Preoperative Assessment System)	30-day mortality + morbidity clusters	8 preoperative predictor variables
NELA risk model (National Emergency Laparotomy Audit)	30-day mortality, emergency laparotomy	Physiology + operative variables; outperforms P-POSSUM in emergency laparotomy
Nottingham Hip Fracture Score (NHFS)	30-day & 1-year mortality after hip-fracture surgery	Age, sex, comorbidities, AMT, Hb, institution, malignancy (used in elderly)

2. Cardiovascular System

Score	Predicts	Notes
Revised Cardiac Risk Index (RCRI / Lee Index)	Major cardiac events, non-cardiac surgery	6 factors; most widely used
Goldman Cardiac Risk Index (original, 1977)	Perioperative cardiac risk	Largely superseded by RCRI
Detsky Modified Cardiac Risk Index	Cardiac risk	Adds angina/CCS class; superseded
Gupta MICA / NSQIP MICA (Myocardial Infarction or Cardiac Arrest calculator)	Peri-op MI/cardiac arrest	Age, ASA, creatinine, functional status, procedure
AUB-HAS2 Cardiovascular Risk Index	Peri-op cardiac events	Heart disease, HF, age, anaemia, vascular surgery, urgency
Vascular Study Group of New England Cardiac Risk Index (VSG-CRI)	Cardiac events after vascular surgery	Vascular-surgery-specific alternative to RCRI
Duke Activity Status Index (DASI)	30-day death/MI via functional capacity	Validated risk tool — 12-item questionnaire; DASI <34 = increased risk; endorsed by ESC 2022
EuroSCORE & EuroSCORE II	Cardiac-surgery mortality	Additive/logistic; II is current standard
STS Risk Score (Society of Thoracic Surgeons)	Cardiac-surgery mortality & morbidity	Procedure-specific models
ACEF / ACEF II (Age, Creatinine, Ejection Fraction)	Cardiac-surgery mortality (simple)	3–4 variables
Revised/other: Kaplan-Feinstein, cardiac index scores	Comorbidity-cardiac risk	Historical

3. Respiratory / Pulmonary System

Score	Predicts	Notes
ARISCAT Score (Assess Respiratory Risk in Surgical Patients in Catalonia)	Postoperative pulmonary complications (PPCs)	7 factors; age, SpO ₂ , resp. infection, anaemia, incision, duration, urgency
Gupta Postoperative Respiratory Failure calculator	Postop respiratory failure (ventilation >48h/reintubation)	NSQIP-derived
Gupta Postoperative Pneumonia calculator	Postop pneumonia	NSQIP-derived
Arozullah Respiratory Failure Index & Arozullah Postoperative Pneumonia Risk Index	Respiratory failure / pneumonia	VA-derived multifactor
Melbourne Group Scale (MGS)	Postop pulmonary complication diagnosis/screening	Clinical/physiological criteria
LAS VEGAS risk score	PPCs	Large intraoperative-practice cohort model
SPORC-1 / SPORC-2 (Score for Prediction of Postoperative Respiratory Complications)	Early postoperative reintubation / respiratory complications	Pre-op (SPORC-1) + intra-op variables (SPORC-2)
STOP-BANG Questionnaire	Obstructive sleep apnoea risk	Snoring, Tiredness, Observed apnoea, Pressure, BMI, Age, Neck, Gender
P-SAP (Perioperative Sleep Apnea Prediction score)	OSA diagnosis	Demographics + history + airway measures
DES-OSA	OSA severity (morphological)	Anatomical/morphometric variables; high specificity
OSA50 / SPOSA / NoSAS	OSA screening	Alternative preoperative OSA screens
ASA OSA checklist	Perioperative OSA risk	Society-of-Anesthesiologists screening checklist
Berlin Questionnaire	OSA risk	3-category symptom tool
Epworth Sleepiness Scale (ESS)	Daytime sleepiness (OSA adjunct)	8-item
STOP Questionnaire / Perioperative STOP-Bang	OSA screening	Shorter STOP form

4. Airway Assessment

Score / Test	Predicts	Notes
Modified Mallampati Test (MMT, Samsoon & Young)	Difficult laryngoscopy/intubation	Class I-IV oropharyngeal view
Cormack-Lehane Grading	Laryngoscopic view difficulty	Grade 1-4 (intra-op)
Wilson Risk Score	Difficult intubation	Weight, head/neck movement, jaw, receding mandible, teeth
El-Ganzouri Risk Index (EGRI / simplified airway risk index)	Difficult intubation	7 weighted variables
LEMON assessment	Difficult airway (emergency)	Look, Evaluate 3-3-2, Mallampati, Obstruction, Neck mobility
Upper Lip Bite Test (ULBT)	Difficult intubation	Class I-III
Thyromental distance (Patil), Sternomental distance, Inter-incisor gap	Difficult intubation	Metric bedside tests
Naguib / combined multivariate airway scores	Difficult intubation	Composite models
El-Orbany / difficult mask ventilation predictors (OBESE, Han grade)	Difficult mask ventilation	Risk-factor sets

5. Hepatic System

Score	Predicts	Notes
Child-Turcotte-Pugh (CTP)	Perioperative mortality in cirrhosis	Bilirubin, albumin, INR, ascites, encephalopathy
MELD (Model for End-Stage Liver Disease)	Mortality in liver disease/surgery	Bilirubin, INR, creatinine
MELD-Na / MELD 3.0	Refined mortality prediction	Adds sodium (and sex/albumin in 3.0)
Mayo Postoperative Mortality Risk (Mayo model)	Post-surgical mortality in cirrhosis	MELD + ASA + age + aetiology
VOCAL-Penn Score	Post-surgical mortality/decompensation in cirrhosis	Contemporary multivariable model
ALBI Grade (Albumin-Bilirubin)	Post-hepatectomy liver failure & mortality (esp. HCC/liver resection)	Albumin + bilirubin only (objective)

6. Renal System

Score	Predicts	Notes
Cleveland Clinic AKI Score (Thakar score)	Postop AKI needing dialysis (cardiac surgery)	Weighted comorbidity model
Mehta / STS AKI models	Dialysis-requiring AKI after cardiac surgery	Multivariable
Simplified Renal Index (SRI)	AKI after cardiac surgery	Reduced-variable alternative to Cleveland/Thakar
KDIGO criteria / AKIN / RIFLE	AKI staging (peri-op)	Creatinine + urine output
eGFR (CKD-EPI/MDRD) & CKD staging	Baseline renal risk	Creatinine-based estimate

7. Nutritional / Metabolic Screening

Score	Predicts	Notes
NUTRIC Score	Nutrition-related ICU/critical-illness risk	Illness severity + nutrition markers
MUST (Malnutrition Universal Screening Tool) / NRS-2002	Malnutrition risk	BMI, weight loss, acute illness

8. Postoperative Nausea & Vomiting (PONV)

Score	Predicts	Notes
Apfel Simplified Risk Score	Adult PONV	4 factors: female sex, non-smoker, PONV/motion-sickness history, postop opioids (0-4) — the standard tool
Koivuranta Score	Adult PONV	5 factors (adds surgery duration ≥ 60 min)
Palazzo & Evans score	PONV	Older multivariable model

(Paediatric PONV — the POVOC / Eberhart score — is under Part B, Paediatric.)

9. Venous Thromboembolism (VTE)

Score	Predicts	Notes
Caprini Risk Score	Surgical VTE risk	Most widely used surgical VTE model; risk rises steeply with score band
Rogers Score	Surgical VTE risk	NSQIP-derived surgical VTE model
Padua Prediction Score	VTE risk (medical inpatients)	Used for medical rather than surgical patients

PART B — CLASSIFIED BY AGE GROUP / SPECIAL POPULATION

10. Paediatric

Score	Predicts / Assesses	Notes
ASA-PS (paediatric application)	Baseline risk	Still the default paediatric tool
APRICOT framework (Anaesthesia PRactice In Children Observational Trial)	Severe perioperative critical events	European multicentre dataset; risk-factor basis, not a single bedside score
Pediatric Risk Assessment (PRAM) Score	Perioperative mortality, non-cardiac surgery	Externally validated risk score
NARCO-SS (Neurological, Airway, Respiratory, Cardiovascular, Other – Surgical Severity)	Paediatric perioperative risk	Organ-system + surgical severity
Pediatric Perioperative Cardiac Arrest (POCA) registry risk factors	Anaesthesia-related cardiac arrest	Registry-derived risk categories
RACHS-1 (Risk Adjustment for Congenital Heart Surgery)	Congenital cardiac surgery mortality	Procedure-based categories
STAT / STS-EACTS Mortality Categories	Congenital cardiac surgery mortality	Procedure complexity tiers
Vasoactive-Inotropic Score (VIS)	Postop severity/outcome (paediatric cardiac)	Cumulative inotrope dosing
SPORC-C	Postoperative respiratory complications in infants/children	Paediatric version of SPORC
POVOC / Eberhart Score	Paediatric postoperative vomiting	4 factors: surgery ≥ 30 min, age ≥ 3 yr, strabismus surgery, POV/PONV family history
Pain tools: FLACC, CHEOPS, FPS-R (Wong-Baker)	Postoperative pain	Age-appropriate scales
mYPAS (modified Yale Preoperative Anxiety Scale)	Preoperative anxiety	Behavioural observation
Paediatric difficult-airway predictors	Difficult intubation	Age-adjusted anatomical assessment

11. Geriatric / Elderly & Frailty

Score	Assesses	Notes
Clinical Frailty Scale (CFS, Rockwood)	Frailty severity (1-9)	Widely used peri-op
Fried Frailty Phenotype	Frailty (physical)	5 criteria: weight loss, exhaustion, weakness, slowness, low activity
Frailty Index (Rockwood deficit accumulation)	Frailty	Ratio of accumulated deficits
Modified Frailty Index — mFI-5 and mFI-11	Surgical frailty/outcomes	NSQIP comorbidity-derived
Edmonton Frail Scale	Multidomain frailty	9 domains incl. cognition, TUG
Groningen Frailty Indicator (GFI)	Frailty screening	15-item
Timed Up and Go (TUG)	Mobility/frailty	Objective timed test
G8 Geriatric Screening Tool	Oncogeriatric vulnerability	8-item
Hospital Frailty Risk Score (HFRS)	In-hospital mortality/complications	109 ICD-10 codes; low/intermediate/high bands; strong discrimination in surgical cohorts
FRAIL Scale	Frailty screening	5 items: Fatigue, Resistance, Ambulation, Illnesses, Loss of weight
Comprehensive Geriatric Assessment (CGA)	Multidomain fitness for surgery	Full assessment, not a single score
Emergency Surgery Frailty Index (EmSFI)	Frailty in elderly emergency surgery	Bedside score
Nottingham Hip Fracture Score (NHFS)	Mortality after hip-fracture surgery	See Part A §1 — primarily an elderly-population score

12. Obstetric

Score	Predicts / Assesses	Notes
Obstetric Comorbidity Index (Bateman)	Maternal end-organ morbidity/mortality	Weighted comorbidities
Placenta Accreta Spectrum risk scoring (e.g. PAS/ultrasound scores)	Accreta risk → haemorrhage/anaesthetic planning	Imaging + history
Airway assessment in obstetrics (Mallampati changes in labour)	Difficult obstetric airway	Mallampati worsens intrapartum
CEMACH/aspiration & general obstetric anaesthetic risk factors	Anaesthesia-related maternal risk	Risk-factor frameworks

PART C — VALIDATION IN THE INDIAN POPULATION

The table below reflects published evidence specifically from Indian cohorts/journals. "Formally validated" = at least one dedicated Indian validation/calibration study; "Studied/used" = Indian data exist but limited or with caveats; "No India-specific data found" = widely used globally but no dedicated Indian validation surfaced in this search.

Score	Indian validation status	Notes from Indian studies
POSSUM & P-POSSUM	☐ Formally validated	Predicts Indian surgical morbidity (POSSUM) and mortality (P-POSSUM) well; P-POSSUM over-predicts in very-low-risk, accurate in high-risk. Validated in general, emergency GI, and neurosurgical (craniotomy) cohorts. A local variant (J-POSSUM , Jabalpur) exists.
EuroSCORE II	☐ Formally validated	Multiple Indian cardiac-surgery validations. Good calibration/discrimination overall; under-predicts mortality in high-risk patients; performs well in low/medium risk. Also studied alongside STS score.
Surgical Apgar Score (SAS)	☐ Formally validated	Validated across South Indian general surgery, Whipple/pancreatic cancer, hollow-viscus perforation, gynae-oncology; lower SAS correlates with more complications/mortality.
SORT	☐ Formally validated	Validated in Indian emergency-surgery cohort (substantial agreement, $\kappa \approx 0.61$; outperformed ASA alone).
ACS-NSQIP calculator	☐ Studied/validated	Evaluated in Indian surgical settings with reasonable performance for complications; some studies question universal applicability to Indian case-mix.
RCRI (Lee Index)	⚠ Studied (limited)	Indian data show event rates rising by RCRI class, but in at least one Indian study only history of IHD was an independent predictor — modest local performance; no large dedicated Indian recalibration.
ARISCAT	⚠ Mixed / studied	Two Indian tertiary-centre studies conflict: poor discrimination (AUC ~ 0.57) in a Hyderabad retrospective laparotomy cohort, but good sensitivity/specificity ($\sim 84\%/80\%$) at a Chennai centre. Utility appears setting-dependent.
STOP-BANG	⚠ Studied, modification proposed	Used for OSA prevalence in Indian cohorts; validity "remains unclear," prompting Indian-specific modifications (STOP-BAWG / STOP-BWANG adding waist circumference) to improve prediction.
Modified Mallampati, ULBT, thyromental/sternomental distance	☐ Studied/validated	Multiple Indian prospective studies (e.g. Bhubaneswar) compare these for difficult intubation; MMT and ULBT comparable, combination improves prediction. Robust Indian airway-prediction literature.
Modified Frailty Index / Clinical Frailty Scale	⚠ Studied	Indian prospective study compared MFI, CFS, ASA-PS and ECOG in cancer surgery; higher frailty predicted worse outcomes. A dedicated Indian frailty tool (FAST – Frailty Assessment & Screening Tool) has also been developed and psychometrically validated in older Indians.
Child-Pugh / MELD	☐ Used/validated	Widely used and studied in Indian hepatology/surgical cohorts (long-standing use; part of Indian liver-disease practice).
ASA Physical Status	☐ Universally used	Standard in Indian practice; used as comparator/reference in most Indian validation studies above.
Charlson Comorbidity Index	⚠ Used	Applied in Indian outcome studies; no dedicated Indian recalibration surfaced.
DASI, Gupta MICA, AUB-HAS2, POSPOM, E-PASS	☐ No dedicated Indian validation found	Used clinically/in research but no India-specific validation surfaced in this search.
Apfel PONV score, Caprini VTE score	⚠ Used, not formally recalibrated	Widely applied in Indian anaesthesia/surgical practice; no dedicated Indian recalibration surfaced in this search.
NELA, SRS, S-MPM, SURPAS, NHFS, ALBI, SPORC, HFRS	☐ No dedicated Indian validation found	Internationally validated; India-specific validation not surfaced in this search.
APRICOT / PRAM / NARCO-SS (paediatric)	☐ No India-specific validation found	APRICOT was European; PRAM externally validated elsewhere. Indian paediatric practice largely relies on ASA-PS; dedicated Indian paediatric-score validation is a gap.
Obstetric Comorbidity Index (Bateman)	⚠ Emerging	Limited formal Indian validation of obstetric comorbidity indices to date.

Bottom line on India: The scores with the strongest dedicated Indian validation are **POSSUM/P-POSSUM**, **P-POSSUM's local variants**, **SORT**, **Surgical Apgar Score**, **EuroSCORE II**, and the bedside airway

tests (Mallampati/ULBT/thyromental distance), plus routine use of **ASA-PS, Child-Pugh and MELD. ARISCAT, STOP-BANG and RCRI** have Indian data but show inconsistent or modest local performance, and Indian-specific modifications have been proposed. **Paediatric perioperative scores remain the biggest validation gap** in Indian data.

Indian validation — sample sizes (N of Indian patients studied)

These are per-study sample sizes from Indian validation studies. Most are single-centre and modest in size; several scores have more than one Indian study, so a range/list is given. "N = 0" means no dedicated Indian validation study was found — the score may still be used clinically, but it has not been formally validated in an Indian cohort in this search.

Score	Indian patients validated against (N)	Study/setting
EuroSCORE II	250 and 1,000 (two studies)	Indian on-pump CABG/valve cohorts, 2011–2012
ARISCAT	330 (Chennai, prospective) + Hyderabad retrospective cohort (~ 510 = derived estimate, not a stated figure)	Two tertiary-care studies, conflicting results
SORT	245	Saveetha, Chennai — emergency surgery
Surgical Apgar Score	220 (S. India district hospital) + 106 (Tripura, hollow-viscus) + others	Multiple prospective studies
Modified Mallampati / ULBT	225	Bhubaneswar — difficult intubation (10.2% incidence)
POSSUM / P-POSSUM	100 (N. India, main study); also 80, 50, 45 in other GI/emergency studies	Prospective general/GI/emergency surgery
Modified Frailty Index / CFS	308 (major abdominal cancer surgery); plus RCC Thiruvananthapuram cancer-surgery cohort	Prospective oncosurgery studies
STOP-BANG	Range 190–700 per study (surgical cohort 204 ; Delhi community 700 ; Gujarat 368 ; S. India urban 190); Indian meta-analysis pooled 11,009 across 8 studies	Prevalence/screening studies, not formal perioperative validation
ACS-NSQIP calculator	Indian data exist but a clear single validation N was not confirmed in this search	Indian tertiary-care evaluation
RCRI (Lee Index)	Indian study exists; exact N not confirmed here	Non-cardiac surgery
ASA-PS	Not a discrete "validation N" — used universally and as the comparator in most studies above	—
Child-Pugh / MELD	Long-standing Indian clinical use; no single discrete validation-N study	Hepatology/surgical cohorts
All other scores in this document (DASI, Gupta MICA, AUB-HAS2, POSPOM, E-PASS, SRS, S-MPM, SURPAS, NELA, NHFS, ALBI, SPORC, HFRS, Apfel, Caprini, APRICOT, PRAM, NARCO-SS, Bateman OCI, etc.)	N = 0 (no dedicated Indian validation study found)	Internationally validated only

Confidence note: only **SORT (245)** and the **Mallampati/ULBT airway study (225)** were confirmed directly from the primary abstract text. The remaining figures come from search-engine summaries of abstracts and should be treated as approximate pending source-checking. The ARISCAT-Hyderabad "~510" is a derived estimate, and the STOP-BANG figures are prevalence surveys rather than formal validation studies.

Caveat: these figures are the sample sizes reported in the specific studies surfaced by this search; other Indian studies with different N almost certainly exist, and "N = 0" reflects the limits of this search rather than proof that no Indian study exists anywhere.

Key Sources

- POSSUM/P-POSSUM in India: Evaluation of POSSUM and P-POSSUM in the Indian population (PMC); J-POSSUM, N. India (PMC); P-POSSUM craniotomy validation (PMC)
- SORT: Development & validation of SORT (BJS); SORT in Indian emergency surgery (Indian J Surg)
- ACS-NSQIP in India: Is NSQIP calculator valuable in the Indian scenario? (Int Surg J)
- Surgical Apgar Score in India: Predictive value of SAS, South India (ScienceDirect); SAS in hollow viscus perforation, Tripura (PMC); SAS Whipple/pancreatic cancer (Springer)
- RCRI: A look into Lee's score (PubMed); Updated review of RCRI (MDCalc/Scholastica)
- ARISCAT in India: ARISCAT validation after laparotomy, India (PMC); ARISCAT utility, tertiary care India (Ovid/JAPT)
- EuroSCORE II in India: EuroSCORE II validation, Indian cardiac surgery (PMC); EuroSCORE II & STS in Indian patients (PubMed)
- STOP-BANG in India: OSA prevalence with STOP-BANG, India (AJPCR); STOP-BANG vs STOP-BAWG/BWANG, India (ERS)
- Airway tests in India: MMT vs ULBT vs Cormack-Lehane, Bhubaneswar (PMC); ULBT vs MMT (Indian J Clin Anaesth)
- Frailty in India: MFI vs CFS vs ASA vs ECOG in cancer surgery (Indian J Surg Oncol); FAST frailty tool for older Indians (Clin Interv Aging)
- Paediatric: APRICOT study (Lancet Respir Med); PRAm external validation (PubMed)
- DASI / METs / functional capacity: METS study, Wijeyesundera et al., Lancet 2018 (PubMed); 2022 ESC guidelines on non-cardiac surgery (Eur Heart J)
- PONV & VTE: Apfel simplified score (MDCalc); Caprini score scoping review (J Vasc Surg Venous)
- Emergency/surgical mortality: NELA risk model (Br J Anaesth); Surgical Risk Scale (PubMed); S-MPM (Annals of Surgery); SURPAS (PubMed)
- Other organ scores: Nottingham Hip Fracture Score (MDCalc); ALBI grade (MDCalc); SPORC-2 (Anaesthesia); SPORC-C paediatric (Br J Anaesth); Hospital Frailty Risk Score in non-cardiac surgery (PLOS One); DES-OSA / P-SAP comparison (PLOS One)